

Integrating factor

Basic skills

Q1. $\frac{dy}{dx} = xy$
↙ separation

a) $\frac{1}{y} dy = x dx$ so $\int \frac{1}{y} dy = \int x dx$

so $\ln y = \frac{1}{2} x^2 + c$

$y = e^{\frac{1}{2} x^2 + c} = \underbrace{e^c}_{\text{Just a constant}} e^{\frac{1}{2} x^2} = A e^{\frac{1}{2} x^2}$

b) $y(2) = 1$
 $1 = A e^{\frac{1}{2} \times 2^2}$ $1 = A e^2$ $A = e^{-2}$

so $y = e^{-2} e^{\frac{1}{2} x^2} = e^{\frac{1}{2} x^2 - 2}$

c) $\frac{dy}{dx} - xy = 0$ so $P(x) = -x$
 $Q(x) = 0$

d) $I = e^{\int P(x) dx} = e^{\int -x dx} = e^{-\frac{1}{2} x^2}$

Then $e^{-\frac{1}{2} x^2} \frac{dy}{dx} - e^{-\frac{1}{2} x^2} xy = 0$

$$\text{so } \frac{d}{dx} (e^{-\frac{1}{2}x^2} y) = 0$$

e) so $e^{-\frac{1}{2}x^2} y = A$ ↙ integrate

$$y = A e^{\frac{1}{2}x^2}$$

Same Answer

Q2. $x^2 \frac{dy}{dx} + e^x y + x = xy$

$$\frac{dy}{dx} + \bar{x}^2 e^x y + \bar{x}' = \bar{x}' y$$

$$\frac{dy}{dx} + (\bar{x}^2 e^x - \bar{x}') y = -\bar{x}'$$

$$P(x) = \bar{x}^2 e^x - \bar{x}' \quad Q(x) = -\bar{x}'$$

Competency skills

Q3. a) $y = 3e^{-5x} + 10x$

$$\text{so } \frac{dy}{dx} = -15e^{-5x} + 10$$

so $\frac{dy}{dx} + sy = -15e^{sx} + 10 + 5(3e^{sx} + 10x)$
 $= 10 + 50x = 10(1+5x)$

b) $\frac{dy}{dx} + sy = 10(1+5x)$

$I = e^{\int s dx} = e^{sx}$ multiply by I

so $e^{sx} \frac{dy}{dx} + se^{sx}y = 10e^{sx}(1+5x)$

$\frac{d}{dx} \left(\underbrace{e^{sx}y}_{Iy} \right) = 10e^{sx} + 50xe^{sx}$

so $e^{sx}y = \int 10e^{sx} + 50xe^{sx} dx$

$y = e^{-sx} \left(2e^{sx} + 50 \left(\frac{1}{5}xe^{sx} - \int \frac{1}{5}e^{sx} dx \right) \right)$ by parts $u=x, dv=e^{sx}$

$= e^{-sx} \left(\cancel{2e^{sx}} + 10xe^{sx} - \cancel{2e^{sx}} + c \right)$

so $y = 10x + ce^{-5x}$

for some constant c

was 3 in part a)

Q4. a) $\frac{dy}{dx} + \frac{2}{x}y = \frac{5}{x^3}$

$$I = e^{\int \frac{2}{x} dx} = e^{2 \ln x} = e^{\ln x^2} = x^2$$

so $x^2 \frac{dy}{dx} + 2xy = \frac{5}{x}$

$$\frac{d}{dx}(x^2 y) = \frac{5}{x}$$

$$x^2 y = \int \frac{5}{x} dx$$

so $y = \frac{1}{x^2} (5 \ln x + c) = \frac{5}{x^2} \ln x + \frac{c}{x^2}$

to verify $\frac{dy}{dx} = -\frac{10}{x^3} \ln x + \frac{5}{x^3} - \frac{2c}{x^3}$

Then

$$\frac{dy}{dx} + \frac{2}{x}y = -\frac{10}{x^3} \ln x + \frac{5}{x^3} - \frac{2c}{x^3}$$

$$+ \frac{2}{x} \left(\frac{5}{x^2} \ln x + \frac{c}{x^2} \right)$$

$$= \frac{5}{x^3} \text{ as required so checked}$$

b) For $\frac{dy}{dx}(1) = 6$ get

$$6 = -\frac{10}{2^3} \ln 2 + \frac{5}{2^3} - \frac{2c}{2^3}$$

$$c = \frac{48 + 10 \ln 2 - 5}{2}$$

$$= \frac{43}{2} + 5 \ln 2$$

$$\begin{aligned} \text{so } y &= \frac{5}{x^2} \ln x + \frac{\frac{43}{2} + 5 \ln 2}{x^2} \\ &= \frac{53}{2x^2} + \frac{5 \ln(2x)}{x^2} \end{aligned}$$

Q5.

$$\cos x \frac{dy}{dx} + y \sin x = x$$

$$\frac{dy}{dx} + \tan x y = \frac{e^x}{\cos x}$$

$$I = e^{\int \tan x dx} = e^{-\ln \cos x} = e^{\ln \sec x}$$
$$\int \frac{\sin x}{\cos x} = - \int \frac{-\sin x}{\cos x} = -\ln \cos x$$

$f'(x)$ points to $\sin x$
 $f(x)$ points to $\cos x$

So $I = \sec x$

$$\sec x \frac{dy}{dx} + \sec x \tan x y = x \sec^2 x$$

$$\frac{d}{dx} (\sec x y) = x \sec^2 x$$

$$\sec x y = \int \underbrace{x \sec^2 x}_{\text{parts}} dx$$

$$= x \tan x - \int \tan x dx$$

$$= x \tan x - \ln \cos x + C$$

$$y = x \cos x \tan x - \cos x \ln \cos x + C \cos x$$

$$= x \sin x - \cos x \ln \cos x + C \cos x$$

Now $y(0) = 1$

So $1 = 0 - 0 \ln 1 + C \cos 0$

$$1 = C$$

So $y = x \sin x - \cos x \ln \cos x + \cos x$

Q6. $u = \frac{y}{x} \quad \frac{dy}{dx} - \frac{y}{x} = \frac{x}{y}$

$y = ux \quad \frac{dy}{dx} = \frac{d}{dx}(ux) = x \frac{du}{dx} + u$

So $x \frac{du}{dx} + u - u = u^{-1}$

So $\frac{du}{dx} = \frac{1}{ux}$

now $\int u \, du = \int \frac{1}{x} \, dx$

$$\frac{1}{2} u^2 = \ln x + C$$

↓ $k=2C$

$$u^2 = 2 \ln x + k$$

$$u = \sqrt{2 \ln x + k}$$

so $\frac{y}{x} = \sqrt{2 \ln x + k} \Rightarrow y = x \sqrt{2 \ln x + k}$

Q7. $y = \frac{1}{u} \quad \frac{dy}{dx} - \frac{y}{x} = x^6 y^2$

$u = \frac{1}{y} \quad \frac{dy}{dx} = \frac{d}{dx} (u^{-1}) = -u^{-2} \frac{du}{dx}$

Chain

$$-u^{-2} \frac{du}{dx} - \frac{1}{ux} = \frac{x^6}{u^2}$$

$$- \frac{du}{dx} - \frac{u}{x} = x^6$$

so $\frac{du}{dx} + \frac{1}{x} u = x^6$

notice
cannot
use Int
factor
as not linear
in u

now $I = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$

$$x \frac{du}{dx} + u = x^7$$

$$\frac{d}{dx}(xu) = x^7$$

$$xu = \int x^7 dx = \frac{1}{8}x^8 + C$$

$$u = \frac{1}{8}x^8 + C$$

$$\frac{1}{y} = \frac{1}{8}x^8 + C \quad y = \frac{1}{\frac{1}{8}x^8 + C} = \frac{8}{x^8 + k}$$

Advanced skills

Q8. $\frac{d}{dx}(I(x)y(x)) = \frac{dI}{dx}y + I \frac{dy}{dx}$

Product rule

now $\frac{dI}{dx} = \frac{d}{dx}(e^{\int P(x) dx}) = \frac{d}{dx}(e^{u(x)})$

Chain

so $\frac{dI}{dx} = u'(x) e^{u(x)}$

where $u(x) = \int p(x) dx$

so $u'(x) = p(x)$

so $\frac{dI}{dx} = p(x) I(x)$

so $\frac{d}{dx} (Iy) = p(x) Iy + I \frac{dy}{dx}$

Q9. $u = \frac{1}{y}$
 $y = \frac{1}{u}$

$\frac{dy}{dx} + (y + x^2 y)^2 = \frac{y}{x}$

$\frac{dy}{dx} = \frac{d}{dx} (u^{-1}) = -u^{-2} \frac{du}{dx}$
chain

$-u^{-2} \frac{du}{dx} + y^2 (1+x^2)^2 = \frac{y}{x}$

$$- \bar{u}^2 \frac{d\bar{u}}{dx} + \bar{u}^2 (1+x^2)^2 = \frac{\bar{u}^{-1}}{x}$$

$$- \frac{du}{dx} + (1+x^2)^2 = \frac{1}{x} u$$

$$\frac{du}{dx} + \frac{1}{x} u = (1+x^2)^2$$

$$I = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$$

$$x \frac{du}{dx} + u = x(1+x^2)^2$$

$$\begin{aligned} \frac{d}{dx} (xu) &= x(1+2x^2+x^4) \\ &= x + 2x^3 + x^5 \end{aligned}$$

$$\begin{aligned} xu &= \int x + 2x^3 + x^5 dx \\ &= \frac{1}{2}x^2 + \frac{1}{2}x^4 + \frac{1}{6}x^6 + C \end{aligned}$$

$$u = \frac{1}{2}x + \frac{1}{2}x^3 + \frac{1}{6}x^5 + \frac{C}{x}$$

$$\text{so } \frac{1}{y} = \frac{1}{2}x + \frac{1}{2}x^3 + \frac{1}{6}x^5 + \frac{c}{x}$$

$$y = \frac{1}{\frac{1}{2}x + \frac{1}{2}x^3 + \frac{1}{6}x^5 + \frac{c}{x}}$$

$$= \frac{6x}{3x + 3x^3 + x^5 + k}$$

Q10. a) $(1-x) \frac{dy}{dx} + \frac{y}{1+x} = (1-x)^2$

$$\frac{dy}{dx} + \frac{1}{(1-x)(1+x)} y = 1-x$$

$$\frac{dy}{dx} + \frac{1}{1-x^2} y = 1-x$$

$$I = e^{\int \frac{1}{1-x^2} dx}$$

$$\text{For } \int \frac{1}{1-x^2} dx$$

use partial fractions

$$\frac{1}{1-x^2} = \frac{A}{1-x} + \frac{B}{1+x}$$

EXAM PAPERS PRACTICE

$$(A-B)x + A+B = 1$$

$$\text{SO } A-B=0$$

$$A+B=1$$

$$A=B=\frac{1}{2}$$

$$\text{SO } \int \frac{1}{1-x^2} dx = \frac{1}{2} \int \frac{1}{1-x} dx + \frac{1}{2} \int \frac{1}{1+x} dx$$

$$= -\frac{1}{2} \ln(1-x) + \frac{1}{2} \ln(1+x)$$

$$= \ln(1+x)^{\frac{1}{2}} + \ln(1-x)^{-\frac{1}{2}}$$

$$\text{SO } I = e^{\ln(1+x)^{\frac{1}{2}} + \ln(1-x)^{-\frac{1}{2}}}$$

$$= (1+x)^{\frac{1}{2}} (1-x)^{-\frac{1}{2}} = \frac{\sqrt{1+x}}{\sqrt{1-x}}$$

$$\begin{aligned} \text{NOW } \frac{d}{dx} \left(\sqrt{\frac{1+x}{1-x}} y \right) &= \sqrt{\frac{1+x}{1-x}} (1-x) \\ &= \sqrt{(1+x)(1-x)} \\ &= \sqrt{1-x^2} \end{aligned}$$

EXAM PAPERS PRACTICE

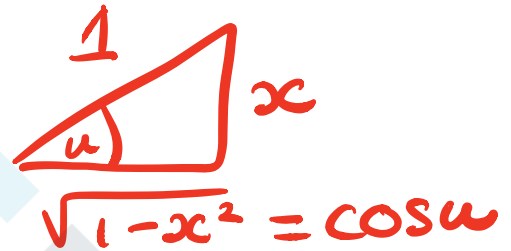
$$\text{So } \sqrt{\frac{1+x}{1-x}} y = \int \sqrt{1-x^2} dx$$

$$\int \sqrt{1-x^2} dx$$

$$\text{Let } x = \sin u$$

$$\text{So } 1-x^2 = \cos^2 u$$

$$\text{and } dx = \cos u du$$



$$\int \sqrt{1-x^2} dx = \int \cos u \times \cos u du$$

$$= \int \cos^2 u du = \int \frac{1}{2} + \frac{1}{2} \cos 2u du$$

double angle

$$= \frac{1}{2} u + \frac{1}{4} \sin 2u + C$$

$$= \frac{1}{2} \sin^{-1}(x) + \frac{1}{4} \times 2 \sin u \cos u + C$$

$$= \frac{1}{2} \sin^{-1}(x) + \frac{1}{2} x \sqrt{1-x^2} + C$$

$$\text{So } y = \sqrt{\frac{1-x}{1+x}} \left(\frac{1}{2} \sin^{-1}(x) + \frac{1}{2} x \sqrt{1-x^2} + C \right)$$

$$y = \frac{1}{2} \sqrt{\frac{1-x}{1+x}} \sin^{-1}(x) + \frac{1}{2} x \sqrt{(1-x)^2} + C \sqrt{\frac{1-x}{1+x}}$$

$$= \frac{1}{2} \sqrt{\frac{1-x}{1+x}} \sin^{-1}(x) + \frac{1}{2} x(1-x) + C \sqrt{\frac{1-x}{1+x}}$$

b) Assuming $1+x > 0$ & $1-x > 0$
 $x > -1$ & $x < 1$

i.e. $|x| < 1$

Strict as have divisions

c) As $y \rightarrow -1$

we have

$$\frac{1}{2} \frac{\pi}{2} \sqrt{\frac{1-x}{1+x}} + -1 + C \sqrt{\frac{1-x}{1+x}}$$

Now if

$$C = -\frac{\pi}{4} \text{ we get } y \rightarrow -1$$

Otherwise $y \rightarrow \pm \infty$